

SRI VASAVI ENGINEERING COLLEGE (Autonomous)

B. Tech V Semester Regular Examinations November 2025

DEEP LEARNING

(Common to CSE(AI) & AIML)

Time: 3 Hrs

Max. Marks:70

PART-A Answer All the Questions.

20 M

1.
 - i. Define McCulloch–Pitts neuron.
 - ii. Describe Biological Neuron.
 - iii. State gradient descent algorithm.
 - iv. Define Multilayer Perceptron (MLP).
 - v. Define the Adam optimization algorithm.
 - vi. Name the full form of RMSProp and Adagrad.
 - vii. State any two advantages of LSTM over RNN.
 - viii. Describe about Pooling in CNNs.
 - ix. Describe Encoder only model of transformers.
 - x. Define self-attention mechanism.

CO1-K1(2M)
CO1-K1(2M)
CO2-K1(2M)
CO2-K1(2M)
CO3-K1(2M)
CO3-K1(2M)
CO4-K1(2M)
CO4-K1(2M)
CO5-K1(2M)
CO5-K1(2M)

PART-B Answer All the Questions.

10 M

2.
 - A.
 - i. Explain the working of McCulloch–Pitts neuron model with an example.
 - ii. Illustrate why a perceptron fails to classify non-linearly separable data with example.
 - OR
 - B.
 - i. Explain the concept of threshold logic and its role in classification.
 - ii. Illustrate commonly used three activation functions and their influence on convergence.
3.
 - A.
 - i. Explain how gradient descent is used for training neural networks.
 - ii. Illustrate the steps of the backpropagation algorithm.
 - OR
 - B.
 - i. Explain the concept and architecture of Autoencoders in deep learning.
 - ii. Explain deep feed-forward neural networks with suitable examples.
4.
 - A.
 - i. Explain the need for optimization algorithms in neural networks.
 - ii. Differentiate first-order and second-order optimization methods in terms of computational complexity and convergence.
 - OR
 - B.
 - i. Explain the effect of learning rate hyper parameter on training speed and accuracy.
 - ii. Explain the batch normalization regularization method for model stability and convergence.
5.
 - A.
 - i. Explain the architecture of RNNs.
 - ii. Explain the architecture and functioning of LSTM cells.
 - OR
 - B.
 - i. Differentiate CNN architectures, AlexNet and ZFNet, highlighting their features.
 - ii. Explain the role of bidirectional RNNs in sequence processing.
6.
 - A.
 - i. Explain the encoder–decoder architecture of Transformers highlighting attention flow.
 - ii. Illustrate the positional encoding applied in the Transformers and its requirement.
 - OR
 - B.
 - i. Explain the working of Variational Autoencoders (VAEs).
 - ii. Illustrate how the GPT models are applied to NLP tasks.

10 M
CO4- K2(5M)
CO4- K2(5M)

CO4- K2(5M)
CO4- K2(5M)

10 M
CO5- K2(5M)

CO5- K3(5M)

CO5- K2(5M)
CO5- K3(5M)

$$\sigma_{t+1} = \sigma_t - \eta \nabla_{\sigma} L_{\sigma t} \quad \nabla$$